



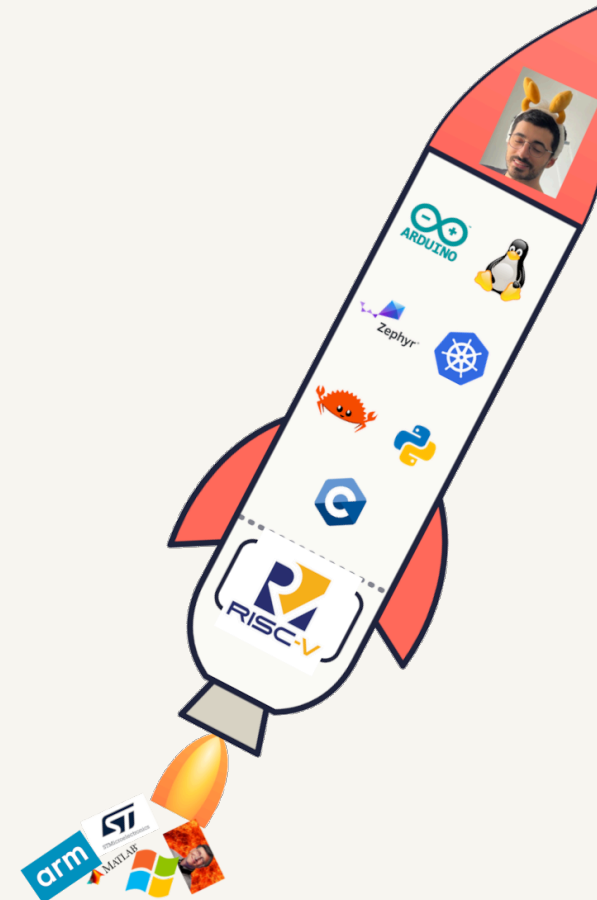
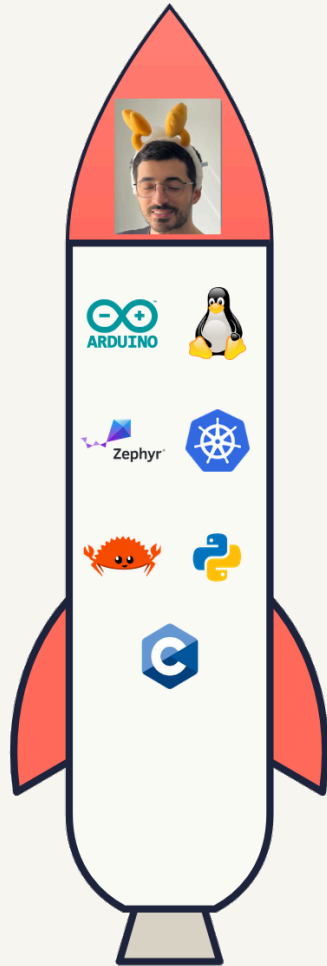
RISC-V Ottawa

Community status update

Inaugural meeting · June 17, 2026 · Yusef Karim

My journey

RISC-V Ottawa



RISC-V is already here

RISC-V Ottawa

Industry

- **NVIDIA** over 1 billion RISC-V cores shipped in their products
- **Meta** RISC-V in their custom AI chips
- **Tenstorrent** designing their own RISC-V AI chips in Toronto
- **Google** every new Chromebook ships with a RISC-V security core
- **Western Digital** RISC-V cores in SSD and HDD controllers
- **Espressif** ESP32-C3/C6/H2, everywhere in IoT and maker products
- **Alibaba** XuanTie RISC-V cores, widely deployed and open-sourced

And spreading fast

- Automotive
- Embedded systems and dev boards
- Space and defence
- A huge and growing open source research ecosystem

Hardware...is still hard, but the barrier to entry has never been lowered.

RISC-V Ottawa exists to **grow** and **consolidate** a **local community** of engineers, researchers, and students in Canada's capital region to **learn, tinker, build**, and, ultimately, **contribute** together across the full **RISC-V** stack: ISA specifications, digital design and verification, FPGAs, embedded systems, operating systems, and applications...nothing is off the table.

Currently about 6 people and a website...

What the group is for

Making individual projects involving RISC-V technologies tractable. Work that is hard to do alone becomes doable via shared knowledge, peer review, and eventually hardware/tooling.

What members might work on

- Implementation on FPGAs
- Verification (UVM, cocotb)
- Learning and testing new ideas in simulation (Verilator)
- Bring-up and drivers for new RISC-V SBCs
- Embedded systems applications (firmware and/or Linux)
- Improving support in firmware/RTOS frameworks
- Upstream toolchain work (GCC, LLVM, QEMU)
- Custom core or ISA extensions

What the group provides

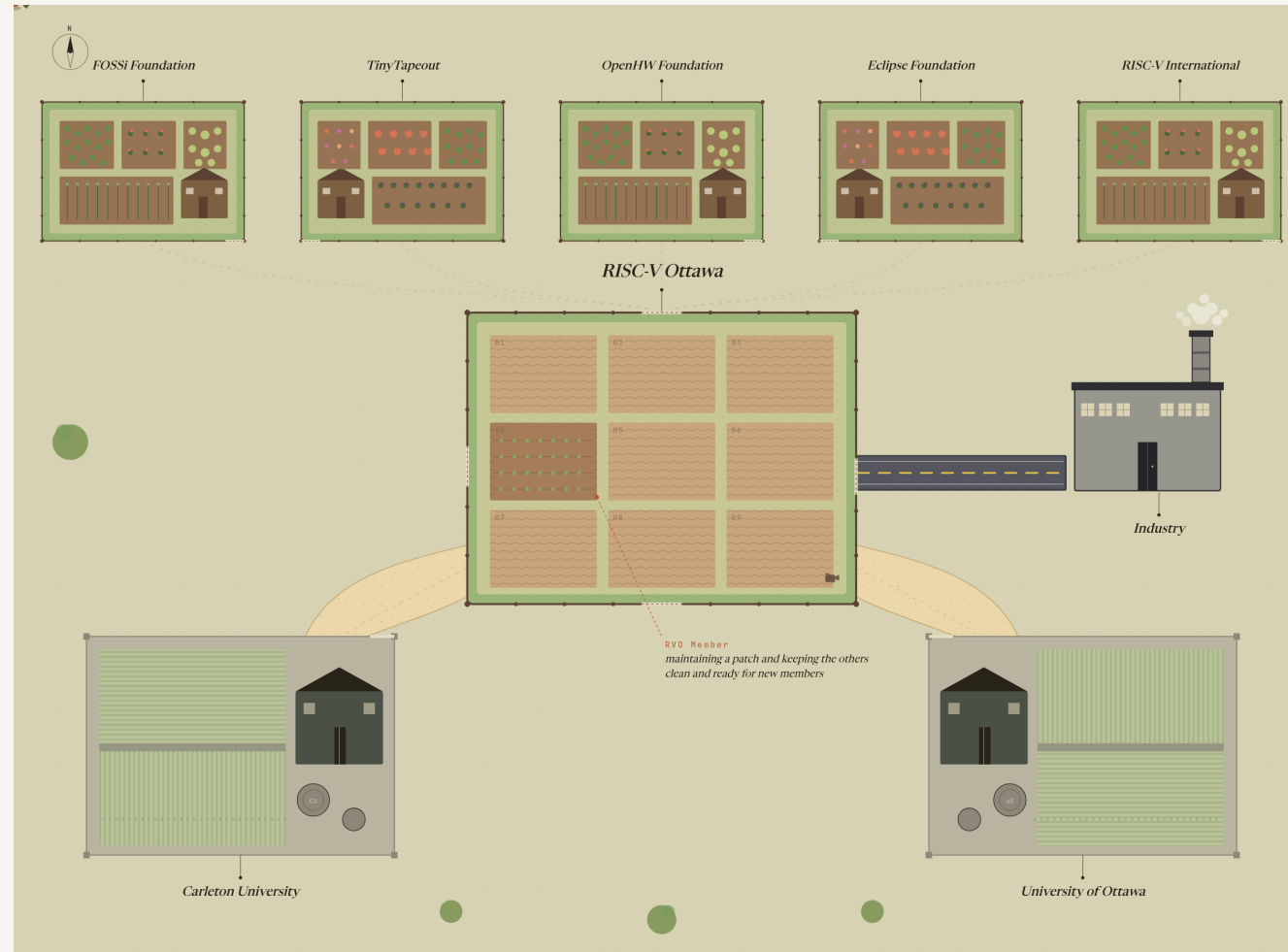
- **Technical**
 - Projects sharing, hands-on hardware session, guided hackathons
- **Educational**
 - Public training (already started), workshops, study groups, university collaborations
- **Social/professional**
 - Networking events, invited speakers

Key point: We will not do this all at once. Instead, we will weave between the technical, educational, and social as opportunities rise and according to the interests of the group.

What's in place

- Website: riscvottawa.ca
 - Curated resource index: riscvottawa.ca/resources
- GitHub: [riscv-ottawa](https://github.com/riscv-ottawa)
- Discord: [RISC-V Ottawa](#)
- Training material (will discuss later)
- We're working on making relationships with others in this area
 - **OpenHW** and **Eclipse** foundations, both presented tonight (thank you ❤️)
 - Actively working on collaborations with **UOttawa** and **Carleton**
 - Actively communicating with other RISC-V communities, e.g., [Works with RISC-V](#)
 - Works with RISC-V is one of the largest “unofficial” RISC-V communities
 - Similar idea as RVO, but global - we will certainly collaborate in the near future

A metaphor



RISC-V Embedded Systems Training **VEGA** edition

Free and open. Bare-metal firmware on the VEGAboard (RV32M1, RI5CY core, RV32IMC). Runs on hardware or entirely in Renode.

1 Hello, firmware world — GPIO, linker script, GDB, Renode

3 Interrupts and timers — trap model, EVENT_UNIT/INTMUX, LPTMR

2 Talking to the world — LPUART0, a console REPL, CSR counters

4  RTOS / Zephyr — west, devicetree, threads

vega.riscvottawa.ca Discussion and help: **Discord #vega-training**

Live demo: VegaConsole

A UART REPL with two interrupt sources running alongside it. Demonstrates timers, interrupts, and how they decouple background work from the foreground loop.

- Heartbeat LED toggled from the LPTMR ISR at 1 Hz; the REPL loop stays in the foreground
- slow busy-loops 2s: the REPL stalls, the heartbeat keeps blinking
- crash executes an illegal instruction; the trap handler logs mcause/mepc and resumes
- SW2 button is a second interrupt (direct EVENT_UNIT line, software debounced) that pauses/resumes the heartbeat

```
vega> crash
about to do something illegal...
trap! cause=2 epc=0x00000abc
...and back!
vega> ticks
g_ticks=412 ( 4 s since boot)
vega>
```

Where we're going

- **Member projects begin:** the first pitches are up next
- **Monthly meetup:** ideally at least one invited speaker (hardware or software), project check-ins, new pitches
- **Study groups** eventually...whenever [RISC-V System-on-Chip Design](#) comes out

Jul 15

(tentative)

meetup # 2

Aug 19

(tentative)

meetup # 3

Sep 16

(tentative)

meetup # 4

Long-term...you decide!

RISC-V Ottawa

★ OPEN-HARDWARE BRIGADE

OTTAWA DIVISION

WE NEED
YOU

CONDITIONS OF ENLISTMENT

// NO CLEARANCE REQUIRED

☐ No tolerance for licence agreements

☐ The will to learn

☐ The patience to read a public spec

Zero royalties. Open future.
Fight for the right to view
source code.



ENLIST AT [RISCVOTTAWA.CA](https://riscvottawa.ca)

DISCORD.GG/EFRYE4WFK4

FORM RV32-I • EST. 2026

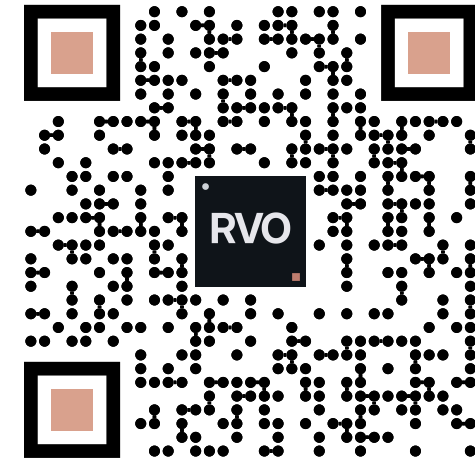
Getting started

RISC-V Ottawa

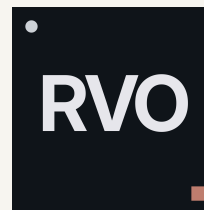
Now Discord, **scan the QR ->**

Summer Pitch your own project, join current projects, read the training, next meetup July 15

Help run it Organizers, a venue, sponsors, university contacts, tell your boss



discord.gg/EfryE4wfk4



riscvottawa.ca · github.com/riscv-ottawa · discord.gg/EfryE4wfk4